OpenDRIVE® 1.8.1, Static Road Network Description, 2024-11-21

13.2 Repeating objects

To avoid lengthy XML code, objects of the same type may be repeated. The attributes of the repeated object may be changed. This element is mainly used to describe railings, railing posts, and street lamps.

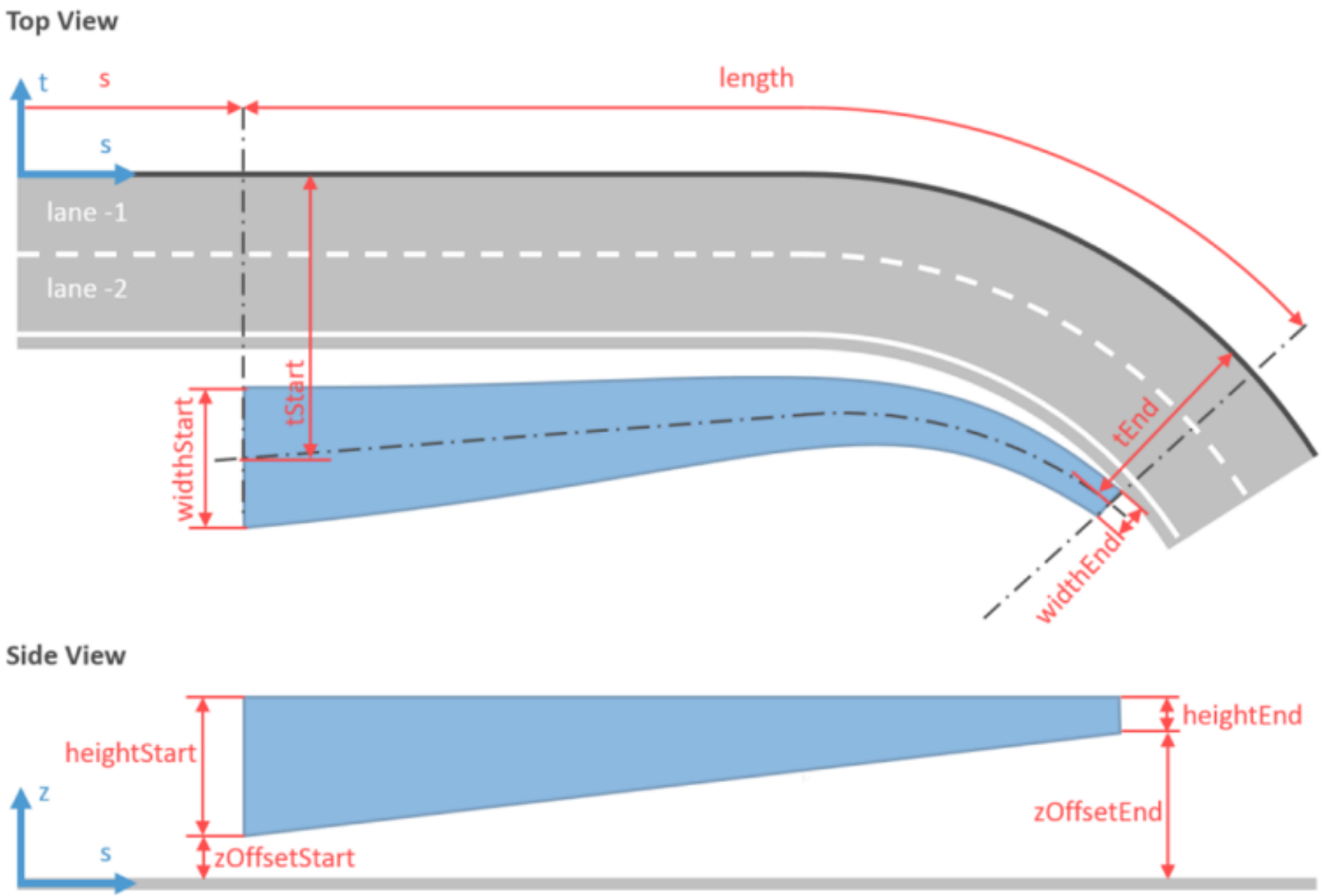

Figure 111. Repeated large angular object

Figure 111 shows top and side views of a large angular object that repeats one other object.

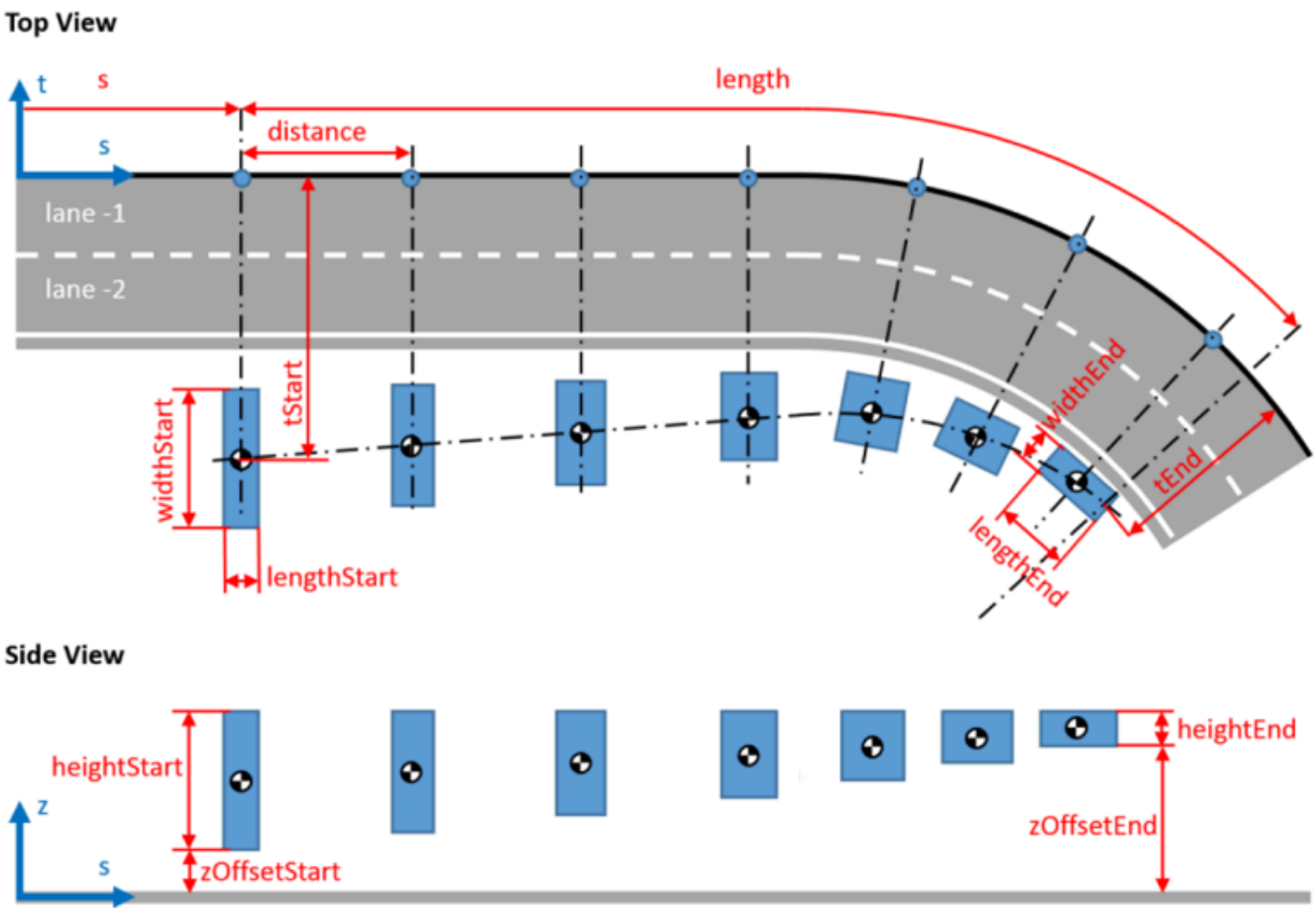

Figure 112. Repeated small angular objects

Figure 112 shows top and side views of several smaller angular objects that are repeated.

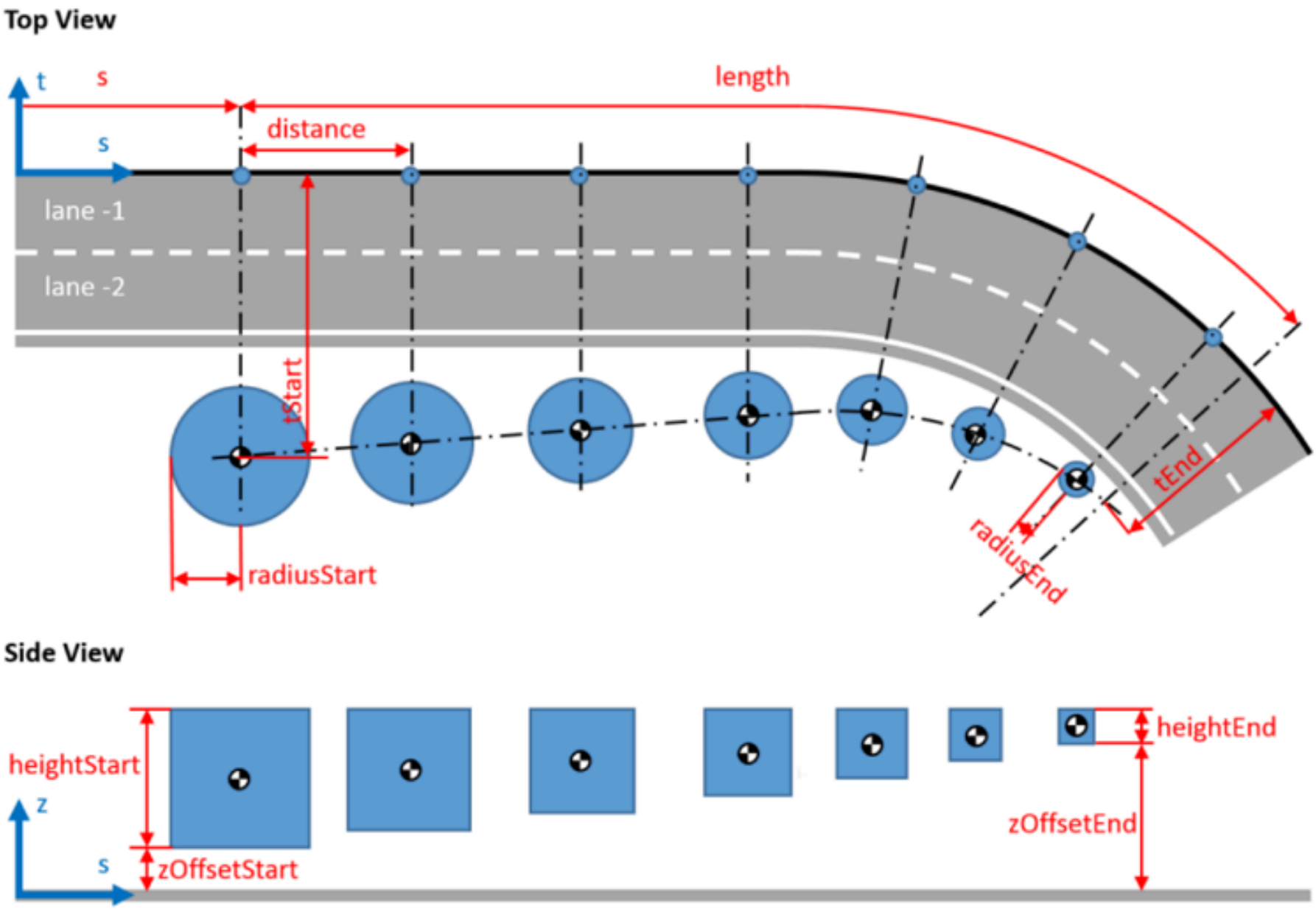

Figure 113. Repeated small circular objects

Figure 113 shows top and side views of several smaller circular objects that are repeated.

Elements in UML model

`<repeat>` element

In ASAM OpenDRIVE, repeating objects are represented by the `<repeat>` element within the `<object>` element.

```
UML class: t_road_objects_object_repeat
XML tag:   <repeat> (Multiplicity: 0..*)
```

To avoid lengthy XML code, objects of the same type may be repeated. Attributes of the repeated object shall overrule the attributes from the original object. If attributes are omitted in the repeated objects, the attributes from the original object apply.

Table 87. Attributes of the <repeat> element

Name	Type	Use	Unit	Introduced	Description
`detachFromReferenceLine`	t_bool	optional		1.8.0	If true, the start and end positions are connected as a straight line which does not follow the road reference line. The default is false
`distance`	t_grEqZero	required	m		Distance between two instances of the object; If this value is zero, then the object is treated like a continuous feature, for example, a guard rail, a wall, etc.
`heightEnd`	t_grEqZero	required	m		Height of the object at @s + @length
`heightStart`	t_grEqZero	required	m		Height of the object at @s
`lengthEnd`	t_grEqZero	optional	m		Length of the object at @s + @length
`lengthStart`	t_grEqZero	optional	m		Length of the object at @s
`length`	t_grEqZero	required			Length of the repeat area, along the road reference line in s-direction.
`radiusEnd`	t_grEqZero	optional	m		Radius of the object at @s + @length

Name	Type	Use	Unit	Introduced	Description
`radiusStart`	t_grEqZero	optional	m		Radius of the object at @s
`s`	t_grEqZero	required	m		s-coordinate of start position, overrides the corresponding argument in the original `<object>` record
`tEnd`	double	required	m		Lateral offset of object's reference point at @s + @length
`tStart`	double	required	m		Lateral offset of objects reference point at @s
`widthEnd`	t_grEqZero	optional	m		Width of the object at @s + @length
`widthStart`	t_grEqZero	optional	m		Width of the object at @s
`zOffsetEnd`	double	required	m		z-offset of the object at @s + @length, relative to the elevation of the road reference line
`zOffsetStart`	double	required	m		z-offset of the object at @s, relative to the elevation of the road reference line

XML example

```xml
<objects>
    <object type="streetLamp"
            name="ExampleStreetLamp"
            id="2"
            s="15.00"
            t="5.0"
            zOffset="0.0"
            orientation="none"
            length="0.14"
            width="1.28"
            height="7.35"
            hdg="0.0"
            pitch="0.00"
            roll="0.0000">
        <repeat s="15.0"
                length="180.0"
                distance="60.00"
                tStart="5.0"
                tEnd="5.0"
                widthStart="1.28"
                widthEnd="1.28"
                heightStart="7.35"
                heightEnd="7.35"
                zOffsetStart="0.0"
                zOffsetEnd="0.0"/>
    </object>
</objects>
```

```
3
2
4
2
5
2
6
2
7
```

Rules

The following rules apply to repeating objects:

- Parameters of the repeated object may differ from the original object.
- Parameters of the repeated object shall overrule the parameters from the original object.
- Repeated objects shall have valid s-coordinates and lengths.

Related topics

- Section 12.13, "Junction groups"
- Section 13.1, "Introduction to objects"